Nicholas Henry

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EDUCATION

Northeastern University Boston, MA
Bachelor of Science in Mechanical Engineering and Computer Science *May 2029*
Coursework: Diff Eq and Lin Alg, Discrete Structures, Program Design and Implementation, Statics *GPA: **3.79***
Activities: Northeastern Electric Racing (FSAE), Students for the Exploration and Development of Space, Deans List
Semester Abroad at Queen's University Belfast in Northern Ireland

SKILLS

Languages: Python, Java, C#, JavaScript, SQL, HTML/CSS, TypeScript, C++
Frameworks & Libraries: React, Next.js, Node.js, Tensorflow, NumPy, Flask, Dash, Tailwind, PostgreSQL
Tools & Platforms: Git, Docker, Github Actions, Redis, Supabase, Linux, ROS, CI/CD, Claude Code, Raspberry Pi, SolidWorks, MATLAB
Certifications & Training: IBM Artificial Intelligence Fundamentals, Microsoft C# Certification, GitHub Foundations

EXPERIENCE

Northeastern Electric Racing - *Python, Dash, React, Docker, Git, Redis, Grafana* Boston, MA
Lead Software Engineer for Performance Analysis Team Jan 2026 – Present
- Develop a parametric vehicle simulation platform (Python backend and React/Dash frontend) used by 120+ engineers across 8 subteams to evaluate how geometry and setup changes effect lap times on custom built tracks
- Pioneered the quasi-steady-state evaluator that calculates maximum cornering speed per track segment, writing results to Redis with a 24 hour TTL so users can visit and compare recent runs without re-simulating
- Lead 15 developers through weekly meetings, review 6+ PRs, and validate the sim using Grafana telemetry to 3% error

Students for the Exploration and Development of Space - *Python, ROS, Git, Linux* Boston, MA
Software Engineer for Network Exploration Swarm Toolkit Team Jan 2026 – Sep 2026
- Program for a team to create a swarm of C.O.B.R.A (Crater Observing Bio-inspired Rolling Articulator) snake robots to explore and develop multi-agent collaboration in order to complete teamwork based exploration tasks
- Created the camera sensor ecosystem consisting of a simultaneous mapping program for multi-agent systems

SEONeeded - *Wordpress, SEOMoz, Semrush* Miami, FL
Web Optimization Engineer Oct 2025 – Aug 2026
- Performed client-facing tasks like identifying keywords, website maintenance, and strategies to target in order to drive more traffic for certain services increasing click-on rate and sales for 15 clients across the Greater South Florida region
- Adapted to the new wave of generative engine optimization space by learning what makes LLM models reference client's websites and how certain worded prompts get tailored responses, recommendations, and links to sources

Search Engine Optimization Intern Jun 2025 – Sep 2025
- Drove a 54% increase in website traffic through targeted optimization by leveraging WordPress and keyword research

PROJECTS

SheepFinder - *TypeScript, React, Next.js, PostgreSQL, Tailwind* May 2026 - Present
- Launched a lost animal reporting app based off studying abroad in Northern Ireland, designing the structured marking system (type, color, and location) based on how farmers identify sheep markings between farms and towns
- Designed a weighted matching engine in TypeScript that scores sightings against registered animals by species, color, and markings and ranking into strong, likely, and possible confidence tiers before notifying users
- Reunited 10+ lost animals with their owners after designing the app around a focus group with 3 farm owners

Portfolio Website - *HTML/CSS, JavaScript, PostgreSQL, Supabase, GitHub Actions* Mar 2025 - Present
- Built a PostgreSQL database with Supabase to have a global comment section where visitors can write something also implementing row-level security to hard lock insert-only access and a database level filter for vulgar text
- Automated résumé updating with a GitHub Actions workflow that fetches it every 6 hours and commits only when it differs from the repo's copy

Text Transcriber - *Python, TensorFlow, Google APIs* Jul 2025
- Trained a CNN with TensorFlow on 100,000+ handwritten character images and obtained 96% validation accuracy
- Tested writing and signage with a group of visually impaired participants to reduce caretaker dependency; results from the limited cohort were an 88% rate of understanding the written text which greatly improved independence